

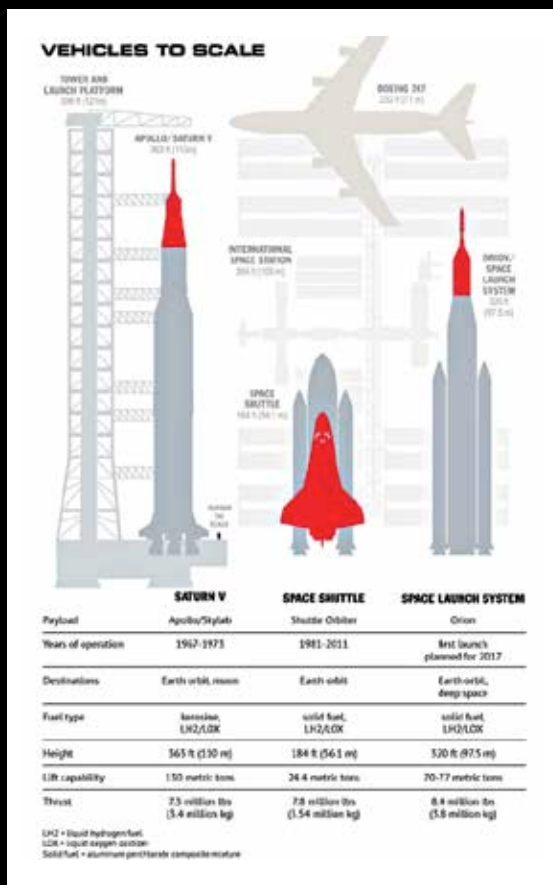


To the moon

Lunar Mission One, a crowdfunded project on Kickstarter, plans on sending a probe down to the moon's surface in 10 years <http://dgit.in/CrowdMoon>

LOTR anyone?

With this new Chrome experiment, you could experience a virtual journey across Middle Earth as well as re-enact epic battles <http://dgit.in/MidEarth>



All current spacecrafts consume up to 90 percent of their launch weight as fuel, making space travel very expensive

In light of this massive restriction, the need to invent a more cost-efficient manner to power our rocket thrusters is critical. This is why NASA's 'Anomalous Thrust Production' paper is rocking the boat of conventional knowledge. As per the experiment conducted by scientists at NASA's advanced propulsion think tank (known as 'Eagleworks Laboratory'), a mechanism has been discovered for generating thrust without the use of any fuel propellant. The technique in question tested the existing idea of a Cannae drive, invented by Guido P. Fetta in a controlled laboratory setting. The experiment involved placing the Cannae drive devices on a super-sensitive torsion pendulum to measure the conversion of electrical power into thrust which moved the pendulum.

Cannae devices are metallic, fully enclosed, conically-tapered cavity, in which one side is narrowly tapered forming a chamber. Inside is a magnetron,

which produces microwaves that cause directional thrust towards the narrower end of the tapering device. The entire process requires no reaction fuel and is powered by electric current. In the case of the NASA experiment, scientists were able to use 30 watts of power to generate between 30-50 micronewtons of thrust – they termed this thrust 'anomalous'. In all practicality, the thrust generated was negligible but points towards a seemingly valid proof-of-concept for the use of such devices for propulsion. The reason this experiment is creating controversy is simply because the 'anomalous thrust' is anomalous and shatters the very foundations of basic physics.

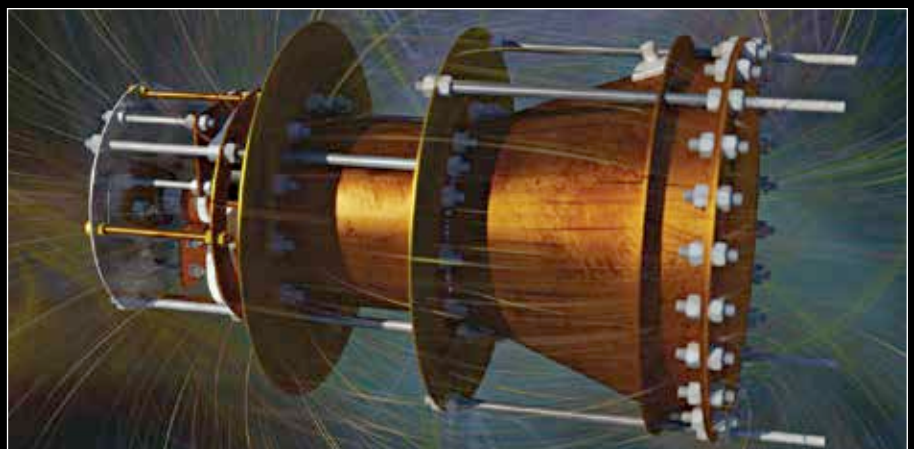
Breaking laws ain't easy

The basic rules of physics and energy are taught in middle-school, but here's a refresher. We all recall that in the natural world there's always conservation of momentum. In physics, momentum is the product of a body's mass and velocity. For centuries, it has demonstrated the phenomena of conversion rather than creation or destruction. The textbook definition of 'conservation of momentum (and energy)' states that neither can be created nor destroyed, simply

converted from one form to another. The results of the NASA experiment fly in the face of this law since the thrust generated is produced as if from thin air. Conventional propulsion takes place when hot gases released on burning fuel are allowed to escape through nozzles against the direction of travel to generate thrust. This is the result of momentum transfer between the escaping gas and the moving craft. However, in the case of NASA's experiment there's no such transference taking place. The equation appears out of balance.

The reactionless propulsive momentum observed in NASA's experiment is vaguely identified as being due to transfer via the "quantum vacuum virtual plasma", which some experts don't even believe to exist. And while NASA remains silent about the detailed physics behind the effects observed, it has said enough to set off both, supporters and admonishers in equal measure. The experiment implies that there's some sort of loophole in conventional physics that's being attributed to "quantum vacuum virtual plasma" which is code for "we don't know what the heck is going on".

Although science thoroughly approves the breaking of old laws with the discovery of new ones, in the case of quantum mechanics the lines of old and new are blurred. The limited observational data on quantum mechanics is significantly limited and even more sparingly understood by scientists. Therefore, it's seen as a lazy answer for phenomena which aren't directly understood under conventional



The NASA-tested Cannae drive is the third such "successful" experiment in reactionless drive technology